

DECODED EXPLAINED

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SIMILIPAL'S BLACK TIGERS ARE GETTING HELP

SPOTLIGHT

Satyasundar Barik

Prakash Chand Gogineni, Field Director of the Similipal Tiger Reserve in Odisha, is still coming to terms with the thrill of a moment he witnessed on May 28. A camera trap had captured a rare sight of a tigress named Zeenat, gently carrying her four cubs one by one in her mouth. Gogineni saw the images in his modest office in Baripada, the district headquarters of Mayurbhanj, where Similipal is located. He knew immediately that he was experiencing a key moment in India's wildlife history.

Zeenat is one of the two tigers introduced into Similipal from Maharashtra's Tadoba-Andhari Tiger Reserve through translocation in 2024. "Similipal was heading towards a genetic collapse, largely due to inbreeding among big cats. While the tiger population was stagnating, the rising number of pseudo-melanistic tigers in the reserve had become a matter of concern," recalls Odisha's former Principal Chief Conservator of Forests (Wildlife) Susanta Nanda.

Pseudo-melanistic tigers (often called black tigers) are a rare variant of the Bengal tiger, distinguished by a genetic mutation that causes their black stripes to widen and merge, giving them a dramatically dark coat where slivers of golden yellow barely peek through.

It was during Nanda's tenure that the two tigresses were introduced to Similipal. He says Zeenat's translocation to Similipal was a necessity, not an experiment. "Maharashtra agreed to part with nine tigers (seven females and two males) in three phases over five years," he says.

Currently, Gogineni says his priority is the safety of the tigress and her cubs. "This can be seen as India's first successful inter-State tiger translocation from a reserve with a surplus population to one with low density," says the 48-year-old Indian Forest Service (IFS) officer, who has been tracking Zeenat for 20 months.

Genetic decoding

This is Odisha's second attempt at translocation – a process of bringing tigers into a new habitat. In 2018, the State brought in two tigers from Madhya Pradesh's Kanha Tiger Reserve and

Similipal's black tigers are getting help

Zeenat, a tigress translocated from Maharashtra's Tadoba-Andhari Tiger Reserve to Odisha's Similipal Tiger Reserve, has given birth to four cubs. Those behind the high-stakes conservation effort share their insights



After giving birth to four cubs, Zeenat carries one in her mouth at Similipal Tiger Reserve in Odisha's Mayurbhanj district, where she was translocated to in 2024. SPECIAL ARRANGEMENT

Bandhavgarh Tiger Reserve to its Satkosia Tiger Reserve. Central India tigers were believed to adapt to the terrain and ecosystem of Odisha faster than those of States further away. The operation went haywire. One of the tigers was poached; the second was returned to its home State due to resistance by locals. Earlier, translocations in

India were aimed at reviving tiger population within States. Spearheaded by the National Tiger Conservation Authority (NTCA), Sariska Tiger Reserve in Rajasthan saw India's first successful tiger reintroduction in 2008. In Madhya Pradesh's Panna Tiger Reserve, NTCA and the Wildlife Institute of India (WII) conducted a similar pro-

gramme in 2009. In both cases, the tigers came from within the States.

In 2021, molecular ecologist Uma Ramakrishnan, who is also a professor at the National Centre for Biological Sciences, Bengaluru, and her co-researchers found that Similipal's tiger population remained small and isolated with minimal genetic exchange with oth-

er landscapes.

In her tiger population genetic analysis, presented in PNAS, a multidisciplinary scientific journal established in 1914, she and her team found that approximately 37% of Similipal's tigers were pseudo-melanistic.

"Small and isolated populations have low genetic variation...making them prone to extinction," the paper said. It also makes the population more vulnerable to diseases.

"These factors might also lead to greater genetic load (harmful genes) in the population. The translocation aimed to reverse all of these. For the long-term survival of a species such as the tiger, genetic diversity becomes the bedrock," Gogineni emphasises.

The 2014 All India Tiger Estimation (AITE), a four-yearly survey led by NTCA and WII, recorded a drastic decline in the tiger population in Similipal to just five. Only one of them exhibited the pseudo-melanistic coat pattern, placing its prevalence at 20%. Subsequent AITEs in 2018 and 2022 indicated a recovery in numbers from five to eight and then 16. However, the rise was accompanied by the worrying genetic drift:

pseudo-melanistic tigers increased to 37.5% in 2018 and 43.75% in 2022. The 2023 All Odisha Tiger Estimation, a State-level survey, found that nearly half of Similipal's tiger population displayed pseudo-melanism. The 2024 AITE further raised concern putting the figure at 59.37%.

Nanda says it was not easy to get tigers from other States. "With a comprehensive report detailing pseudo-melanism in tigers, a strong prey base, and an increased patrolling system in Similipal, we approached the Ministry of Environment, Forest and Climate Change. After much persuasion, the Ministry agreed and the approval of the NTCA followed in May 2024," he recalls.

Maharashtra to Odisha

The approval paved the way for the translocation. Jamuna, 2.5 years, was released in October 2024 into a specially designed soft enclosure at a meadow in southern Similipal's Jenabil.

Two weeks later, on November 15, Zeenat, the second tigress from Tadoba-Andhari, then three years old, was released in the early hours into an enclosure at Tiger meadow in Chaha-

la, in northern Similipal.

Jamuna made her way to the Kuldiha forest, which lies outside Similipal's core area. Zeenat, however, traversed three States: Odisha, Jharkhand and West Bengal. This is now regarded as one of India's most intense pursuits of a single tiger, says Nanda.

In April 2025, Zeenat was released into the wild. "The tigress changed her territory twice and finally mated with T45 in January this year and delivered four cubs in May," says Gogineni.

Nanda says India's tiger conservation programme is facing the challenge of habitat connectivity. "In that context, Zeenat broke the barrier of discontinuity in the habitat."

Now 25% of cubs in Similipal have a diverse gene pool and it will grow stronger in three-four years if Zeenat's cubs survive, he adds.

This inter-State tiger translocation programme has achieved its medium-term objective. Now, Odisha awaits the decision of a high-level committee on the translocation of the remaining seven of the nine tigers from Maharashtra.

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SIMILIPAL BLACK TIGERS ARE GETTING HELP

Tiger is the largest cat species in the world. It is both an umbrella species.

Keystone species: A species whose addition to or loss from an ecosystem leads to major changes in the occurrence of at least one other species.

Flagship species: A species chosen to represent an environmental cause, such as an ecosystem in need of conservation.

Subspecies:

There are eight recognised subspecies of tigers, out of which three are extinct.

Bengal tiger: Indian subcontinent

Amur tiger: Amur Rivers region of Russia and China, and North Korea

South China tiger: South central China.

Sumatran tiger: Sumatra, Indonesia.

Indochinese tiger: Continental south-east Asia.

Caspian tiger: Turkey through central and west Asia (extinct).

Javan tiger: Java, Indonesia (extinct).

Bali tiger: Bali, Indonesia (extinct).

SIMILIPAL BLACK TIGERS ARE GETTING HELP

Countries with most tigers:

1st India > 2nd Russia > 3rd Indonesia > 4th Nepal > 5th Thailand

Threats:

Habitat loss, climate change, human-wildlife conflict, **inbreeding**, diseases, hunting, grazing, mining, infrastructure development, tiger farms (big cats are held captive for breeding and sale), etc.

SIMILIPAL BLACK TIGERS ARE GETTING HELP

● The Genetic Problem

- Pseudo-melanistic tigers ('black tigers') are a rare Bengal tiger variant caused by a genetic mutation that widens and merges black stripes, giving a dark coat
- **Similipal small, isolated tiger population was heading towards genetic collapse due to inbreeding**
- **Small, isolated populations have low genetic variation, making them prone to extinction, disease and greater genetic load**

● The Translocation Effort

- After NTCA approval in May 2024, Maharashtra agreed to part with nine tigers (seven females, two males) from Tadoba-Andhari Tiger Reserve in three phases over five years
- Tigress Jamuna was released into a soft enclosure in southern Similipal in October 2024; tigress Zeenat followed in November 2024 in northern Similipal
- This is Odisha's second translocation attempt - the first, from Madhya Pradesh to Satkosia (2018), failed after one tiger was poached and the other returned due to local resistance

SIMILIPAL BLACK TIGERS ARE GETTING HELP

Project Tiger

It is a **Centrally Sponsored Scheme of the Ministry of Environment, Forest and Climate Change** (MoEFCC).

It was launched in **1973 for in-situ conservation** of wild tigers in designated tiger reserves.

The **National Tiger Conservation Authority (NTCA)** has an overarching supervisory role, performing functions as provided under the Wildlife (Protection) Act, 1972.

Global Tiger Forum (GTF)

It is the **only** inter-governmental, international body working exclusively for the conservation of Tigers in the Wild. It is a forum of 13 tiger range countries.

Goal: Doubling wild tigers count by 2022. India achieved this target in 2018, 4 years ahead of deadline.

SIMILIPAL BLACK TIGERS ARE GETTING HELP

TX2 Goal

Target: To double the number of wild tigers.

Conservation Assured | Tiger Standards (CA|TS)

An initiative of the World Wildlife Foundation (WWF), it is a set of criteria that allows tiger sites to check if their management will lead to successful tiger conservation.

Global Tiger Initiative (GTI)

It was launched in 2008 as a global alliance of governments, civil society and conservation and scientific communities to save wild tigers from extinction.

Microsoft introduces pay-as-you-go model for AI agents



John Xavier

For the past two years, the AI industry has been trying to solve a billing problem: how do you charge users for artificial intelligence?

Bundling AI services into software subscriptions and selling it the way enterprise software has been sold for years looked fine for some time. Microsoft, Google and others added AI assistants to their productivity suites and put a monthly price tag on them.

That logic worked if companies were already paying per employee for email, spreadsheets and collaboration tools. But AI is unlike traditional software.

One employee may use an AI assistant a few times a week to summarise meetings. Another may use it all

day to analyse documents, write reports and automate repetitive tasks. Increasingly, AI systems are also becoming agents that can perform actions on behalf of users, carrying out tasks that would previously have required hours of human effort. This leads to a mismatch between how AI is consumed and how it is billed.

That is what makes Microsoft's latest move interesting. The company has introduced a pay-as-you-go model for its AI agents, allowing customers to pay according to usage rather than committing solely to fixed licences.

What Microsoft is really doing is moving AI closer to the economics of cloud computing.

Cloud computing became a trillion-dollar business because it replaced large upfront investments with consumption-based pricing. Companies could rent computing power as required and pay for what



Microsoft has introduced a pay-as-you-go model for its AI agents, allowing customers to pay according to usage. REUTERS

they used.

The model worked because it aligned costs with value. If a business needed more computing resources, it paid more. If demand fell, spending fell too. AI seems to indicate a similar trend.

The challenge with subscription pricing for AI services is that it assumes every user derives roughly the same value from the service. In reality, AI usage varies enormously. Some workers barely touch these tools. Others rely on them constantly. The gap is like-

ly to become even wider as AI agents become more capable and autonomous.

A flat monthly fee begins to look increasingly arbitrary in such a world. So, the new pricing attempts to solve that problem by linking spending to usage. Customers can experiment without making large commitments.

But the reason behind Microsoft taking this route isn't just to give its customers more autonomy.

For instance, once Word is installed, Microsoft does not incur any additional

cost each time a customer types out a new document. The product has been sold. But with AI, the business relationship is back and forth. Every prompt, every response and every agent action consumes computing resources. This type of supplier and customer relationship will shoot up the token generation cost for cloud providers.

No wonder Microsoft picked a consumption-based model that reflects this reality far more accurately than a fixed subscription. In effect, the Windows software maker is likely treating AI as an organism that grows and takes new forms inside an organisation.

The company's experience with Azure suggests it understands how to build operating models around metered consumption. Enterprises have already become comfortable paying for storage, computing power and databases this way. Paying for AI

capacity may not be such a dramatic leap.

Yet there are reasons for caution. The same pricing model that makes cloud computing flexible can also make it unpredictable. Anyone who has ever been surprised by a cloud-services bill knows that usage-based pricing has a downside. Customers like paying only for what they use, but they also like knowing what next month's bill will look like.

But as the use of AI agents expands in organisations, it will be difficult to forecast how much to budget for the next quarter. Organisations face a challenge akin to modernising the plumbing of an old, large building.

If AI agents become deeply embedded in their business processes, usage could rise rapidly. Microsoft's challenge will be convincing customers that they can scale their AI deployments without losing control of spending.

Cloud Computing

Cloud computing distributes computer services such as servers, storage, databases, networking, software, analytics, and intelligence through the Internet (“the cloud”) to enable quicker innovation, more flexible resource allocation, and cost savings.

It **eliminates the need for individuals and businesses** to self-manage physical resources themselves and only pay for what they use.

Public Cloud: Services are delivered over the Internet and accessible to authorised users. The cloud provider manages all hardware and software components. Examples include Amazon Web Services (AWS) and Microsoft Azure.

Private Cloud: Exclusively used by a single organisation and runs on a private network. Access is limited to internal users and can be on-premises or hosted by third-party providers. E.g, VMware Cloud.

Hybrid Cloud: Combines elements of both public and private clouds, allowing for on-premises storage of sensitive data while utilising the public cloud for less sensitive information.

Cloud Computing

Infrastructure as a Service (IaaS) : Provides on-demand access to IT infrastructure services such as compute, storage, networking, and virtualisation. Offers highest control over IT resources, similar to traditional on-premises IT resources.

Platform as a Service (PaaS) : Offers hardware and software resources needed for cloud application development, allowing companies to focus on application development without managing infrastructure.

Software as a Service (SaaS) : Delivers a full application stack as a service, including infrastructure, maintenance, and updates, typically for end-user applications managed by the service provider.

Edge Computing involves decentralized data processing near the source to minimize latency and optimize bandwidth for real-time applications.

Edge computing enables faster and more extensive data processing with low latency and at less cost than cloud computing, resulting in real-time action-led outcomes.

Cloud Computing vs Edge Computing

Feature	Cloud Computing	Edge Computing
Processing Location	Remote data center	Near the data source
Speed	Slower (internet delay)	Faster
Internet Dependency	High	Low
Storage Capacity	Very high	Limited
Privacy	Data sent to cloud	Data stays local
Best For	Big data analytics, storage	Real-time applications

Cloud Computing vs Edge Computing

Which one of the following services does NOT come under category of Cloud computing ?

This question was previously asked in

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1. IaaS (Infrastructure as a Service)
2. SaaS (Software as a Service)
3. PaaS (Platform as a Service)
4. BDaaS (Big data as a Service)

Cloud Computing vs Edge Computing

UPSC Prelims 2022 Science and Tech

Q. With reference to “Software as a Service (SaaS)”, consider the following statements:

1. SaaS buyers **can customise** the user interface and can change data fields.
2. SaaS users **can access** their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

Which of the statements given above are correct?

a 1 and 2 only

b 2 and 3 only

c 1 and 3 only

d 1, 2 and 3

INSURING THE INFORMAL WORKER AGAINST THE SOARING HEAT

Insuring the informal worker against the soaring heat

Jacob Koshy

NEW DELHI

Shabbir Alam works at construction sites in Haryana, though his home is in Ludhiana, Punjab. Last month, ₹1,000 appeared in his bank account. His contractor had enrolled him and some colleagues in a scheme under which, if temperatures crossed a certain threshold, money would be paid. "I earn ₹800-₹1,000 a day from my work... Some of us got ₹1,000 last month, so it was helpful," he told *The Hindu* in a phone conversation.

Harion, who drives a leased taxi in the Delhi-NCR region, received ₹2,000 over two months. "I'm happy that I got this... We anyway have a company rule that we can avoid driving on very hot days from 12 p.m. to 4 p.m., but extra money is always welcome," he told *The Hindu*.

Alam and Harion are among 3,925 informal workers enrolled this year across Delhi-NCR – Noida, Delhi, Gurugram, Ghaziabad and Faridabad – under a "parametric" heat-in-

surance product run by the non-profit Jan Sahas, with Go Digit General Insurance and the CSR arm of Godrej Properties among its partners.

Parametric insurance

Circuiting the need for policyholders to pay premiums or submit claims, parametric insurance depends on a pre-set trigger, calculated by insurers from a historical dataset on temperature, humidity, rainfall and air quality over a region. If a day's weather index crosses a predetermined threshold, or an index, it would trigger a payout to the worker's bank account with no questions asked. Last year, out of roughly 6,000 workers enrolled, 722 in Noida received ₹1,000 when that city's trigger was activated.

According to the IMD's May weather bulletin, only on a single day, maximum temperatures in Delhi were at least 5°C 'above normal' – one of the qualifying criteria to be a 'heat-wave' day. The IMD has forecast 'above-normal' heatwave conditions over much of India in 2026, and



Extreme weather: Workers earning a living outdoors get payouts only after temperatures cross a threshold. KOMMURI SRINIVAS

the World Meteorological Organization (WMO) expects El Niño, which tends to weaken the monsoon and intensify heat, to develop after July, following a summer that pushed temperatures past 47°C in some places.

"Insurance companies usually set heat-index payout thresholds by looking at past weather and loss data to find the point where extreme heat starts causing real problems (using correlation between heat wave, excess rainfall or AQI with health), such as reduced productivity or health risks," Shailesh

Acharya, Director, Jan Sahas, said in an email response through a spokesperson.

Several such schemes now exist. The Self-Employed Women's Association's (SEWA) programme, the largest of its kind, grew from 21,000 women in Gujarat in 2023 to about 2.25 lakh women across seven States by 2025, paying out a few hundred rupees whenever temperatures exceed 40°C, according to a study by the Council for Inclusive Capitalism.

No State government yet runs a heat-insurance scheme for workers – Na-

galand's is the lone government parametric policy that insures residents against excess rainfall. Almost everywhere, premiums are paid by philanthropic organisations rather than workers themselves.

Behavioural change

Whether the money alters behaviour is a more difficult question. In a randomised trial involving 276 gig delivery workers in Delhi and Gurugram, published on the Social Science Repository Network, economists Chen, Y.M.A Hossain and S. Sekhri found that a ₹200 payment at the start of a heatwave let workers shift their schedules to cooler parts of the day and rest. Those who received only heat warnings worked 0.8 fewer days and reported far more headaches and fatigue.

"Our goal is to drive greater accountability among industry and government stakeholders for safeguarding workers against climate-related challenges, including extreme heat and air pollution, which can result in

adverse health outcomes and loss of income," said Mr. Acharya, adding, "Ultimately, workers should have the option of deciding whether they want to continue working under extreme climatic conditions or take time off until conditions become safer and more tolerable, without being forced to choose between their health and their income."

Under the Jan Sahas-led scheme too, workers get payouts only after temperatures actually cross a threshold, and they don't get a specific forecast on whether temperatures are likely to cross a threshold on a particular day.

Jatin Singh, founder of weather forecasting firm Skymet, who has provided data for parametric schemes to cover agricultural losses, said such programmes could work only under government or philanthropic largesse. "Parametric insurance does not have its own legs," Mr. Singh told *The Hindu*.

"Without somebody else paying the premium, the model doesn't work," he said.

INSURING THE INFORMAL WORKER AGAINST THE SOARING HEAT

Heatwave is considered if the maximum temperature of a station reaches at least 40°C or more for Plains and at least 30°C or more for Hilly regions.

Coastal Regions : When the maximum temperature departure is 4.5°C or more from normal, a Heatwave may be described, provided the actual maximum temperature is 37°C or more.

Heat Wave: Departure from normal is 4.5°C to 6.4°C.

Severe Heat Wave: Departure from normal is >6.4°C.

Heatwaves is not a notified disaster in Notified Disaster List

INSURING THE INFORMAL WORKER AGAINST THE SOARING HEAT

● **The Problem**

- Outdoor informal workers - construction labourers, gig delivery workers, taxi drivers - face income and health risks from extreme heat, with no formal safety net
- 3,925 informal workers were enrolled this year in heat-insurance schemes across Delhi-NCR (Noida, Delhi, Gurugram, Ghaziabad, Faridabad)

● **How Parametric Insurance Works**

- Bypasses the need for policyholders to pay premiums or submit claims; payout triggers automatically once a pre-set weather index (temperature, humidity, rainfall, AQI) crosses a threshold
- Last year, of about 6,000 workers enrolled, 722 in Noida received Rs.1,000 when the city's heat trigger was activated

● **Key Schemes and Partners**

- Run by non-profit Jan Sahas with Go Digit General Insurance and Godrej Properties' CSR arm; the SEWA programme, the largest of its kind, grew from 21,000 women in Gujarat (2023) to 2.25 lakh women across seven States (2025)
- **Nagaland is the only State government running a parametric policy, but it covers excess rainfall, not heat; premiums elsewhere are paid almost entirely by philanthropic organisations**

Thank You